

WEEK 3

Advanced Cybersecurity Practical Task

Technical Learning & Security Assessment
Final Report



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Organization
DG Interns Hub

*Stronger Networks.
Safer Tomorrow.*

DOCUMENT MAP

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01

Introduction

From networking fundamentals to an evidence-based security assessment.

Week 3 extended the networking concepts learned earlier into a controlled cybersecurity assessment workflow. A two-VM VMware lab was created so that discovery, service enumeration, packet capture, traffic investigation and hardening could be performed against an authorized target without relying on public systems.

The assessment deliberately separated observation from interpretation. Nmap was used to establish the target surface, Wireshark was used to observe the underlying traffic, and the final findings were based only on evidence produced in the lab.

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Objectives

- Build and validate an isolated VMnet1 cybersecurity lab.
- Discover live hosts and exposed services using Nmap.
- Perform service/version, OS, SYN, UDP and NSE assessments.
- Capture and interpret ICMP, DNS, TCP, UDP and ARP traffic with Wireshark.
- Correlate Nmap port states with packet-level TCP behavior.
- Assess security posture and distinguish confirmed findings from hardening opportunities.
- Apply controlled firewall hardening and validate the change with before/after testing.

Assessment principle

Only the user-owned VMs on the authorized lab network were scanned and analyzed. No public or third-party systems were included.

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Lab Environment

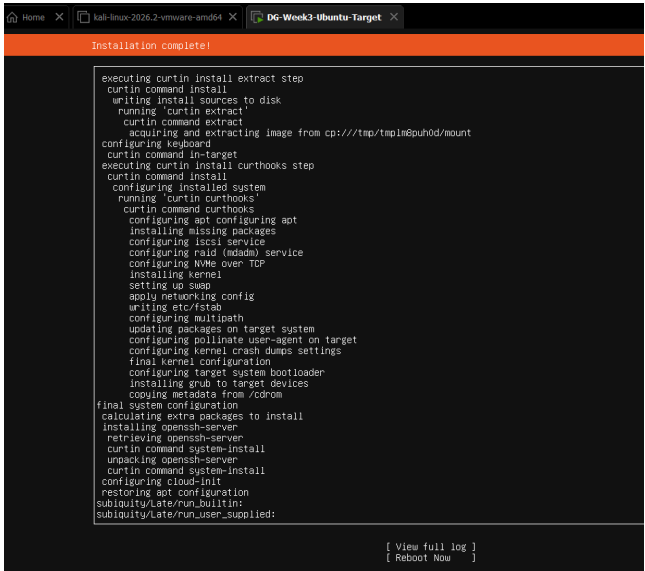
Virtualization, addressing and target preparation.

Component	Configuration
Hypervisor	VMware Workstation
Kali Linux	4 GB RAM, 2 vCPU, eth0 = 192.168.162.128/24
Ubuntu Server Target	2 GB RAM, 2 vCPU, 20 GB disk, hostname ubuntu-target
Primary network	VMnet1 Host-only, 192.168.162.0/24
SSH service	OpenSSH Server installed on Ubuntu
Packet analysis	Wireshark on Kali eth0

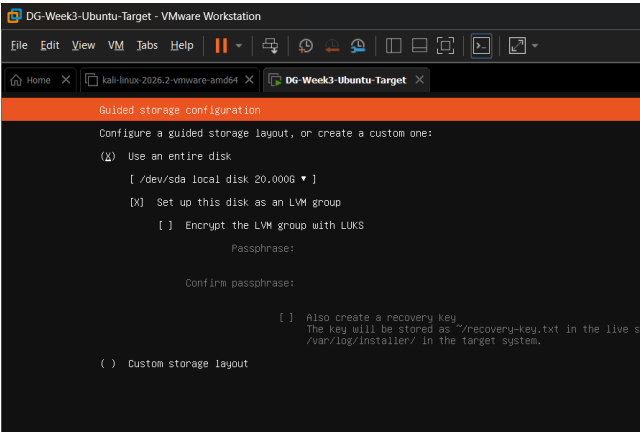
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Installation & Target Preparation

Ubuntu Server was installed as the dedicated target and OpenSSH Server was enabled during installation. Kali was used as the assessment VM. Host-only networking provided the isolated assessment path.



Installation completed with OpenSSH Server.



Target storage configuration.

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Network Diagram

Authorized assessment architecture - primary path is isolated VMnet1.

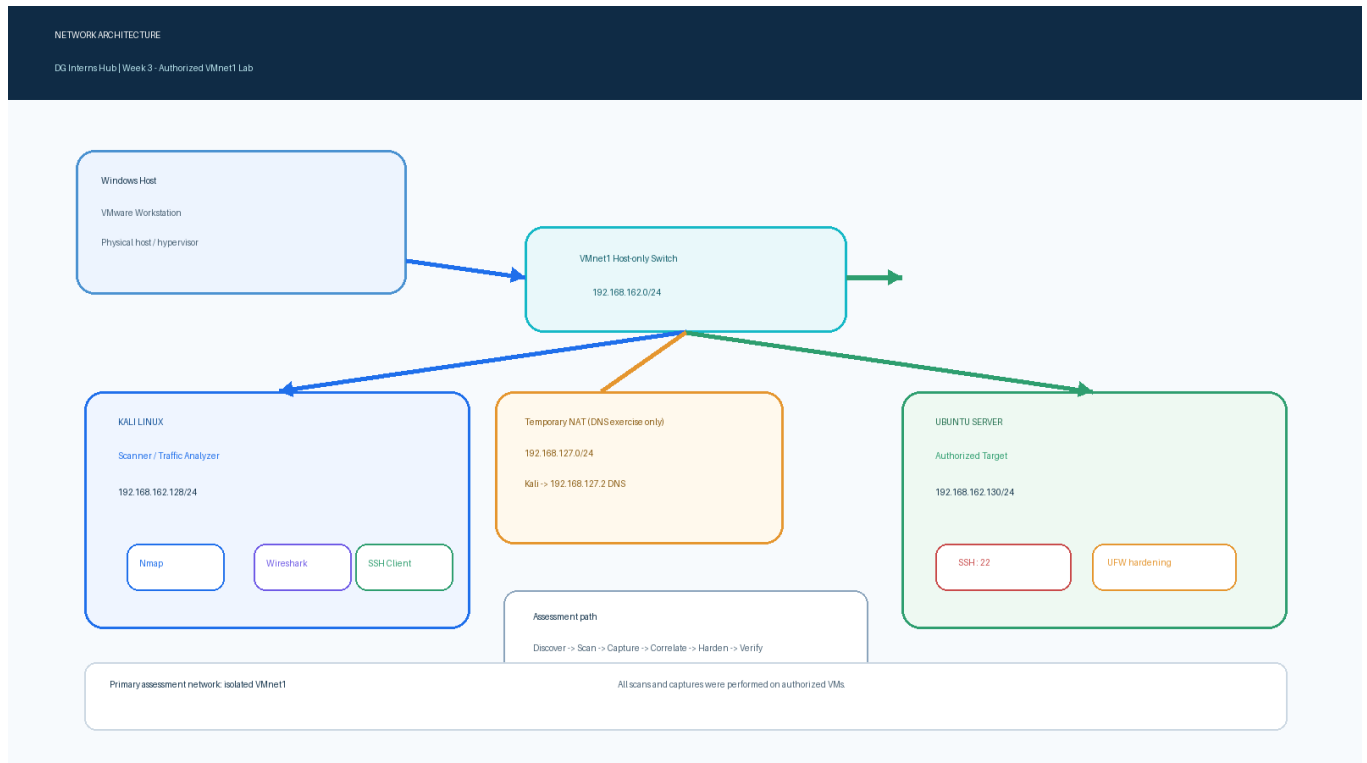


Figure 1. Advanced Week 3 lab topology. The temporary NAT path was used only for the DNS exercise and was not part of the assessment network.

Network roles

Kali = scanner + packet analyzer | Ubuntu = authorized target | VMnet1 = isolated assessment segment | SSH/22 = intentionally exposed service for validation.

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Tools Used

Tools were selected to cover discovery, analysis and validation.

Tool	Purpose
VMware Workstation	Virtualize the lab systems and create the Host-only assessment network.
Kali Linux	Assessment workstation for Nmap, SSH and packet analysis.
Ubuntu Server	Authorized target and hardening platform.
Nmap	Host discovery, TCP/UDP enumeration, service/version and NSE assessment.
Wireshark	Packet capture, protocol filtering and traffic investigation.
UFW	Host-based firewall used for the hardening phase.

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Nmap Methodology

A repeatable assessment sequence.

Stage	Command / focus	Purpose
Discovery	<code>nmap -sn 192.168.162.0/24</code>	Identify responding hosts.
Baseline scan	<code>nmap 192.168.162.130</code>	Identify exposed default TCP ports.
Service detection	<code>nmap -sV 192.168.162.130</code>	Identify service/version details.
OS detection	<code>nmap -O 192.168.162.130</code>	Attempt network fingerprinting.
SYN scan	<code>nmap -sS 192.168.162.130</code>	Observe TCP SYN-based port states.
UDP scan	<code>nmap -sU --top-ports 20 192.168.162.130</code>	Review common UDP exposure.
NSE review	<code>nmap --script vuln 192.168.162.130</code>	Run vulnerability-oriented scripts.

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Nmap Results

Baseline discovery and exposed-service evidence.

Host discovery

The 192.168.162.0/24 lab scan identified four responsive addresses. The relevant assessment machines were Kali 192.168.162.128 and Ubuntu 192.168.162.130; the remaining responsive addresses were VMware virtual-network infrastructure.

Assessment	Observed result
Host discovery	4 hosts up on the /24; Kali and Ubuntu were the relevant VMs.
Basic TCP scan	22/tcp open SSH; 999 default TCP ports closed before UFW hardening.
Service/version	OpenSSH 10.2p1 Ubuntu 2ubuntu3.5 identified on 22/tcp.
OS detection	No exact OS match returned; one-hop network distance.
UDP top 20	No port conclusively identified as open; several reported open filtered.
NSE vuln scripts	No vulnerability findings reported in the scan output.

```

kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nmap -sS -p 22,9998 192.168.162.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 09:05 -0400
Nmap scan report for 192.168.162.130
Host is up (0.0011s latency).
PORT      STATE SERVICE
22/tcp    open  ssh
9998/tcp   closed distinct32
MAC Address: 00:0C:29:76:16:13 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 4.70 seconds
kali@kali:~$

```

Evidence - Controlled Nmap correlation scan: 22/tcp open and 9998/tcp closed.

Advanced Nmap Analysis

Interpreting the results without over-claiming.

Check	Result	Analyst interpretation
-sV	OpenSSH 10.2p1 Ubuntu 2ubuntu3.5	Provides a concrete service/version baseline for later assessment.
-O	No exact OS match	Fingerprinting was inconclusive; the target is known to be Ubuntu from lab configuration.
-sS	22/tcp open	SYN behavior confirmed the open SSH service.
UDP top 20	open filtered on several ports	Not treated as confirmed open services without additional evidence.
--script vuln	No findings reported	Automated result is not proof that the host has zero vulnerabilities.

Key interpretation

Open ports describe attack surface. A service/version match or scanner result is evidence for follow-up, not proof of exploitability by itself.

Wireshark Methodology

Capture first, filter second, interpret with context.

Wireshark was used on Kali eth0 to inspect packet-level behavior on the authorized lab network. Each traffic pattern was generated deliberately, captured, filtered and then interpreted against the known lab topology.

Capture focus	Filter used	Purpose
ICMP	icmp	Show Echo request/reply behavior.
DNS	dns	Observe query and response pairs.
TCP/SSH	tcp.port == 22	Observe TCP handshake and SSH negotiation.
UDP	udp	Observe a controlled UDP datagram.
ARP	arp	Observe IP-to-MAC address resolution.
TCP SYN	tcp.port == 22 && tcp.flags.syn == 1	Isolate SYN/SYN-ACK behavior.
TCP RST	tcp.port == 9998 && tcp.flags.reset == 1	Observe reset behavior for a closed port.

Temporary NAT note

Kali temporarily used NAT for the DNS capture because 192.168.162.1 did not provide a working DNS service. The VM was returned to Host-only / VMnet1 for the security assessment.

Packet Analysis

Selected packet-level evidence from the controlled captures.

No.	Time	Source	Destination	Protocol	Length	Info
13	0.0000000	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request, seq=7502, ttl=64 (request in 10)
14	0.0000000	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) request, seq=7502, ttl=64 (reply in 10)
15	0.0000000	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request, seq=7502, ttl=64 (request in 10)
16	0.0000000	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) request, seq=7502, ttl=64 (reply in 10)
17	0.0000000	192.168.1.100	192.168.1.1	ICMP	60	Echo (ping) request, seq=7502, ttl=64 (request in 10)
18	0.0000000	192.168.1.1	192.168.1.100	ICMP	60	Echo (ping) request, seq=7502, ttl=64 (reply in 10)

ICMP: 4 Echo requests + 4 replies

No.	Time	Source	Destination	Protocol	Length	Info
19	0.0000000	192.168.1.100	192.168.1.1	TCP	60	55555 -> 22 [ACK] Seq=1000000000, Len=0, Win=0, RST=1 (RST in 10)
20	0.0000000	192.168.1.1	192.168.1.100	TCP	60	55555 -> 22 [ACK] Seq=1000000000, Len=0, Win=0, RST=1 (RST in 10)
21	0.0000000	192.168.1.100	192.168.1.1	TCP	60	55555 -> 22 [ACK] Seq=1000000000, Len=0, Win=0, RST=1 (RST in 10)
22	0.0000000	192.168.1.1	192.168.1.100	TCP	60	55555 -> 22 [ACK] Seq=1000000000, Len=0, Win=0, RST=1 (RST in 10)
23	0.0000000	192.168.1.100	192.168.1.1	TCP	60	55555 -> 22 [ACK] Seq=1000000000, Len=0, Win=0, RST=1 (RST in 10)
24	0.0000000	192.168.1.1	192.168.1.100	TCP	60	55555 -> 22 [ACK] Seq=1000000000, Len=0, Win=0, RST=1 (RST in 10)

TCP/SSH: SYN -> SYN-ACK -> ACK + SSHv2

No.	Time	Source	Destination	Protocol	Length	Info
25	0.0000000	192.168.1.100	192.168.1.1	DNS	100	Standard query query, id=1, flags=0x00, op=0, rcode=0, qtype=1, qclass=1
26	0.0000000	192.168.1.1	192.168.1.100	DNS	100	Standard query response, id=1, flags=0x00, op=0, rcode=0, qtype=1, qclass=1
27	0.0000000	192.168.1.100	192.168.1.1	DNS	100	Standard query query, id=1, flags=0x00, op=0, rcode=0, qtype=1, qclass=1
28	0.0000000	192.168.1.1	192.168.1.100	DNS	100	Standard query response, id=1, flags=0x00, op=0, rcode=0, qtype=1, qclass=1
29	0.0000000	192.168.1.100	192.168.1.1	DNS	100	Standard query query, id=1, flags=0x00, op=0, rcode=0, qtype=1, qclass=1
30	0.0000000	192.168.1.1	192.168.1.100	DNS	100	Standard query response, id=1, flags=0x00, op=0, rcode=0, qtype=1, qclass=1

UDP: controlled 33655 -> 9999 datagram

No.	Time	Source	Destination	Protocol	Length	Info
31	0.0000000	192.168.1.100	192.168.1.1	ARP	42	Request, op=1, hwtype=0, ptype=0, hlen=6, plen=6, ethertype=0x0800, protocol=0x0008, sender_haddr=00:00:00:00:00:00, target_haddr=00:00:00:00:00:00
32	0.0000000	192.168.1.1	192.168.1.100	ARP	42	Reply, op=2, hwtype=0, ptype=0, hlen=6, plen=6, ethertype=0x0800, protocol=0x0008, sender_haddr=00:00:00:00:00:00, target_haddr=00:00:00:00:00:00
33	0.0000000	192.168.1.100	192.168.1.1	ARP	42	Request, op=1, hwtype=0, ptype=0, hlen=6, plen=6, ethertype=0x0800, protocol=0x0008, sender_haddr=00:00:00:00:00:00, target_haddr=00:00:00:00:00:00
34	0.0000000	192.168.1.1	192.168.1.100	ARP	42	Reply, op=2, hwtype=0, ptype=0, hlen=6, plen=6, ethertype=0x0800, protocol=0x0008, sender_haddr=00:00:00:00:00:00, target_haddr=00:00:00:00:00:00
35	0.0000000	192.168.1.100	192.168.1.1	ARP	42	Request, op=1, hwtype=0, ptype=0, hlen=6, plen=6, ethertype=0x0800, protocol=0x0008, sender_haddr=00:00:00:00:00:00, target_haddr=00:00:00:00:00:00
36	0.0000000	192.168.1.1	192.168.1.100	ARP	42	Reply, op=2, hwtype=0, ptype=0, hlen=6, plen=6, ethertype=0x0800, protocol=0x0008, sender_haddr=00:00:00:00:00:00, target_haddr=00:00:00:00:00:00

ARP: target IP -> MAC resolution

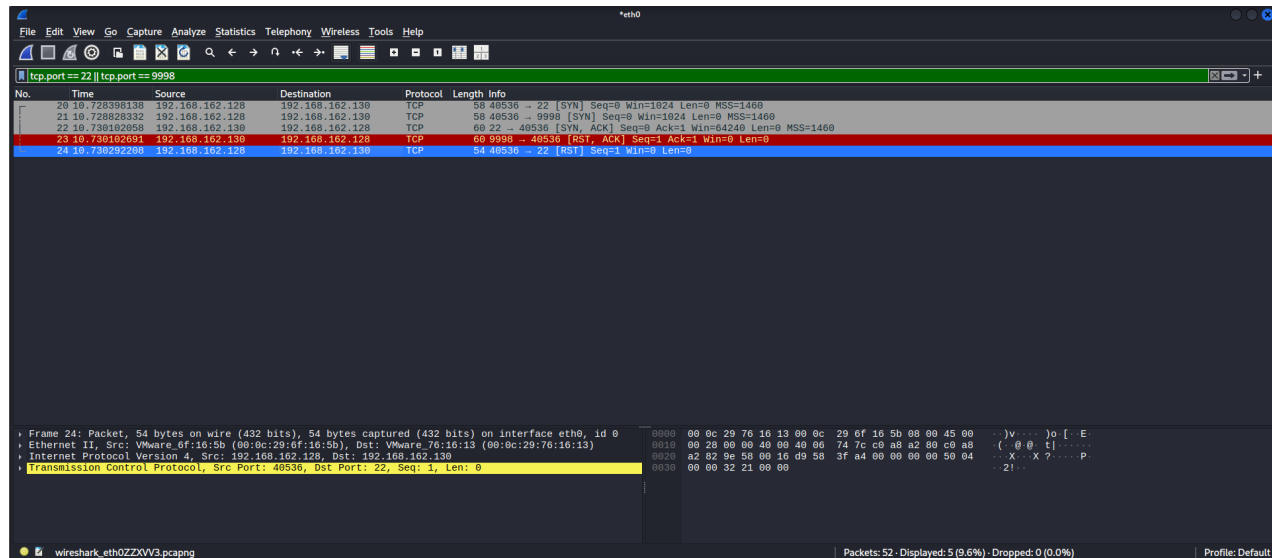
Observed protocols

ICMP, DNS, TCP/SSH, UDP and ARP were observed. The captures provided packet-level evidence for the connectivity, service and network-layer behaviors documented in the assessment.

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Nmap + Wireshark Investigation

Correlating scanner output with TCP packet behavior.



Evidence - Controlled port 22 / 9998 capture during Nmap SYN scan.

Nmap state	Observed packets	Interpretation
22/tcp open	SYN -> SYN-ACK -> ACK	Target accepted the TCP connection attempt; SSH service was reachable.
9998/tcp closed	SYN -> RST/ACK	Target reset the attempt; behavior was consistent with a closed port.

Correlation result

Nmap port-state classifications matched the packet-level TCP responses observed by Wireshark, strengthening the evidence behind the assessment.

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Vulnerability Assessment

Evidence-based review of the target posture.

The vulnerability review combined the Nmap NSE result with direct target checks. No confirmed software vulnerability was established in the collected evidence. Instead, two security-hardening opportunities were identified and validated through configuration inspection.

Finding	Evidence	Assessment
UFW inactive	sudo ufw status verbose -> inactive	Security hardening gap in the pre-hardening baseline.
SSH password authentication enabled	sshd -T showed passwordauthentication yes	Hardening opportunity; not treated as a confirmed vulnerability by itself.
Root password login prohibited	permitrootlogin prohibit-password	Positive control; not a finding.
Public-key authentication enabled	pubkeyauthentication yes	Positive control; not a finding.
No pending updates observed	apt list --upgradable returned no packages	No update finding from this check.
Nmap NSE vuln scan	No findings reported	Automated scan did not report a vulnerability.

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Risk Classification

Severity was assigned to the security posture evidence, not to a speculative exploit.

Item	Classification	Reason
UFW inactive	Medium	Unsolicited inbound traffic was not filtered at the host firewall layer before hardening.
Password authentication enabled	Low / hardening	Acceptable in some environments but weaker than key-only SSH for administrative access.
SSH exposed	Attack surface	Intentional lab service; exposure alone is not a vulnerability.
No NSE findings	Informational	No automated vulnerability finding was produced; validation limits remain.

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Security Hardening

Controlled firewall hardening with validation.

- Recorded pre-hardening state: UFW inactive.
- Recorded the pre-hardening Nmap baseline: 22/tcp open; other default ports closed.
- Allowed SSH before enabling the firewall to preserve authorized administration.
- Enabled UFW with low logging.
- Repeated the same Nmap baseline using -T4.
- Verified SSH remained accessible after hardening.

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Before / After Comparison

The same target and scan were used to make the comparison meaningful.

Metric	Before UFW	After UFW
Firewall status	Inactive	Active
SSH / TCP 22	Open	Open
Other default TCP ports	Closed (RST response)	Filtered (no response)
SSH administrative access	Available	Verified available

Measured effect

The firewall change did not remove the required SSH service, but it changed unsolicited access to the other scanned TCP ports from closed responses to filtered/no-response behavior.

```

Ubuntu 26.04.1 LTS ubuntu-target tty2
Try contacting this VM's SSH server via 'ssh vsock32950804659' from host.
ubuntu-target login: targetadmin
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Sep 20 06:26:30 PM UTC 2026
System load: 0.12          Memory usage: 23%    Processes:   213
Usage of /: 48.5% of 9.75GB Swap usage:   0%    Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

targetadmin@ubuntu-target:~$ sudo ss -ltnp
sudo: authentication required
targetadmin@ubuntu-target:~$ sudo ss -ltnp
NetID  State Recv-Q  Send-Q      Local Address:Port      Peer Address:Port      Process
udp    UNCONN 0        0          127.0.0.1:53             0.0.0.0:*               users:((("chronyd",pid=1250,fds=4))
udp    UNCONN 0        0          127.0.0.54:53            0.0.0.0:*               users:((("systemd-resolve",pid=764,fds=18))
udp    UNCONN 0        0          192.168.162.1:53         0.0.0.0:*               users:((("systemd-resolve",pid=764,fds=26))
udp    UNCONN 0        0          0.0.0.0:53               0.0.0.0:*               users:((("chronyd",pid=1250,fds=5))
tcp    LISTEN 0        4096       127.0.0.1:22             0.0.0.0:*               users:((("chronyd",pid=1250,fds=5))
tcp    LISTEN 0        4096       127.0.0.1:22             0.0.0.0:*               users:((("ssh",pid=1216,fds=3),("systemd",pid=1,fds=189))
tcp    LISTEN 0        4096       127.0.0.1:22             0.0.0.0:*               users:((("systemd-resolve",pid=764,fds=19))
tcp    LISTEN 0        4096       0.0.0.0:22               0.0.0.0:*               users:((("ssh",pid=1216,fds=4),("systemd",pid=1,fds=189))

targetadmin@ubuntu-target:~$ sudo ufw status verbose
Status: inactive
targetadmin@ubuntu-target:~$ sudo sshd -T | grep -E "(permitrootlogin|passwordauthentication|pubkeyauthentication)"
#PermitRootLogin prohibit-password
#PubkeyAuthentication yes
#PermitRootLogin local yes
targetadmin@ubuntu-target:~$ apt list --upgradable
targetadmin@ubuntu-target:~$

```

Evidence - Target baseline checks used to guide the hardening phase.

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Security Recommendations

Practical next steps based on the observed posture.

- Keep UFW enabled and review inbound rules regularly.
- Prefer SSH public-key authentication and disable password authentication when operationally appropriate.
- Limit management access by source network wherever practical.
- Maintain normal Ubuntu and OpenSSH patch management.
- Monitor unusual SYN bursts, RST responses and retransmissions in context.
- Retest the exposed surface after service or configuration changes.

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Challenges Faced

Troubleshooting was treated as part of the assessment workflow.

Challenge	How it was resolved	Learning
SSH initially appeared inactive	Started/verified the service and then validated port 22 from Kali.	Installation and runtime state are separate checks.
IPv4 ICMP was initially missing in Wireshark	Forced IPv4 ping and captured the correct exchange.	Capture timing and protocol version matter.
DNS failed on Host-only network	Temporarily used NAT for DNS capture, then returned to VMnet1.	Lab services and network roles must be verified before capture.
Wireshark filters showed background traffic	Refined filters to target specific ports and flags.	Filtering is essential for evidence quality.

Analyst lesson

Good security reporting is as much about controlled troubleshooting and accurate interpretation as it is about running tools.

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Conclusion

Assessment outcome and final learning.

The Week 3 assessment progressed from lab construction to host discovery, service enumeration, packet capture, traffic investigation, vulnerability review and controlled hardening. Nmap established the target surface, Wireshark provided packet-level evidence, and the final UFW change demonstrated a measurable reduction in unsolicited TCP exposure while preserving the required SSH service.

Key takeaway

The strongest improvement was the ability to connect a tool result to an observable network event, then make a controlled configuration change and prove its effect with a before/after test.

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References

Source	Reference
Nmap Documentation	https://nmap.org/book/man.html
Wireshark Documentation	https://www.wireshark.org/docs/
Ubuntu Server Documentation	https://documentation.ubuntu.com/server/
Ubuntu UFW Documentation	https://documentation.ubuntu.com/server/how-to/security/firewalls/
OpenSSH	https://www.openssh.com/

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Screenshots / Evidence

Selected evidence stored with the submission set.

Evidence set	Examples
Nmap	Network discovery, basic scan, service/version, OS, SYN, UDP and NSE outputs.
Wireshark	ICMP, DNS, TCP/SSH, UDP, ARP, SYN and RST captures.
Investigation	Nmap + Wireshark correlation and retransmission analysis.
Hardening	Target baseline, UFW before/after, and before/after Nmap comparison.